

Wave Equations: Lecture 1

Introduction

Professor Nicholls

Department of Mathematics, Statistics,
and Computer Science
University of Illinois at Chicago

Wave Equations: Lecture 1

Numerical Simulation of Wave Equations

- In this project we will study the **numerical simulation** of **wave equations**.
- Wave equations arise in **all** areas of science and engineering: Acoustics, electromagnetics, seismology, etc.
- While the governing PDEs can vary in their complexity, the principal features of solutions can be observed in the **Wave Equation**

$$u_{tt} = c^2 u_{xx}.$$

- To investigate solutions of this (and other) wave equation we will use the **Finite Difference Method**.
- To verify our **derivations** and **implementations** we will compare to **exact solutions** given by the **Method of Characteristics**.

The Finite Difference Method

- The **Finite Difference Method** is an excellent place to start:
 - It is **easy** to derive schemes.
 - It is **easy** to implement schemes.
 - The resulting implementations are often extremely fast.
 - Their numerical analysis can be accomplished with Calculus.
- However, they have several drawbacks which can make more sophisticated algorithms appealing:
 - They can have a **low order** of accuracy.
 - They can be **dispersive** (speed wrong).
 - They can be **dissipative** (amplitude wrong).
- Philosophy:
 - Subdivide **spatial** interval, $\{0 < x < L\}$, with **nodes** $0 = x_0 < x_1 < \dots < x_{J-1} < x_J = L$.
 - Subdivide **temporal** interval, $\{0 < t < T\}$, with **nodes** $0 = t_0 < t_1 < \dots < t_{N-1} < t_N = T$.
 - **Unknowns:** The values $v_{j,n} \approx u(x_j, t_n)$, the **true solution**.
 - **Equations:** Demand that the PDE, ICs, BCs, be true at the gridpoints (x_j, t_n) .

Chapter 10 of LeVeque's Book

- To get us started I am sending you my lecture notes for **Chapter 10** of *Finite Difference Methods for Ordinary and Partial Differential Equations* by R. LeVeque.
- This book has the websites:
 - <https://faculty.washington.edu/rjl/fdmbook/>
 - <https://epubs.siam.org/doi/book/10.1137/1.9780898717839>
- While the book provides MATLAB scripts, I would encourage you to write your own (e.g., in MATLAB, C, C++, python, julia).
- I would recommend
 - Reading these lecture slides with the book.
 - Implementing all of the algorithms for the **one-way wave equation**

$$u_t + au_x = 0.$$

- Investigate the behavior of these schemes as J and N are **refined**:
Convergence? Stability?

Future Directions

- Study the details of the **Method of Characteristics**.
- Study **nonlinear** wave equations displaying **solitons**, e.g., the Korteweg–de Vries (KdV) Equation

$$u_t + 6uu_x + u_{xxx} = 0.$$

- Study **nonlinear** wave equations displaying **breaking**, e.g., the Burgers Equation

$$u_t + uu_x = 0.$$

- Study the full **Maxwell Equations**.
- Systems of **Hyperbolic Conservation Laws**.
- More careful **Numerical Analysis** found in LeVeque.

Chapter 1: Finite Difference Approximations

- Our goal is to **solve** Ordinary Differential Equations (ODEs) and Partial Differential Equations (PDEs).
- **Why?** Because they arise in engineering and scientific applications of great importance and interest.
- We know from long experience that closed-form “exact solutions” (in terms of the classical Calculus functions) are **impossible** to find save in the simplest cases.
- **One approach:** Find **numerical approximations** which approximate the true solution by some design philosophy.
- **Note:** In this course we will only address **Well-Posed Problems**.
- These are ODEs/PDEs, supplemented with side (Initial/Boundary) conditions for which:
 - A solution **exists**.
 - The solution is **unique**.
 - The solution depends **continuously** on all problem parameters.

Finite Difference Methods

- Any numerical method must specify two design principles:
 - What are the **unknowns**?
 - What **equations** must these unknowns satisfy?
- In the **Finite Difference Method** these are
 - **Unknowns:** The values of the unknown function (dependent variable) at a **finite** number of values of the independent variable(s).
 - **Equations:** Make the ODE/PDE be **true at the finite number of values chosen**.
- This approach has a number of key advantages:
 - Clearly the number of equations equals the number of unknowns which, for linear ODE/PDE, gives us hope for a **unique solution**.
 - These schemes are **simple** to derive and implement!

Right-Hand Finite Difference Approximation

- However, we begin with a simpler task: Approximate the derivative of a function using only a **finite** number of its values.
- For precision, suppose that $u(x)$ is “smooth” (**Applied Math:** It has enough derivatives to justify any formula we write down!) and that the point in question is $\bar{x}$.
- Inspired by Calculus we consider the **right-hand approximation**

$$D_+ u(\bar{x}) := \frac{u(\bar{x} + h) - u(\bar{x})}{h}, \quad h \ll 1.$$

- We know from Calculus that, in the limit $h \rightarrow 0$, this gives the **exact** derivative.
- **Note:** $D_+ u(\bar{x})$ is the **slope** of the line **interpolating** $u(x)$ at the points $\{\bar{x}, \bar{x} + h\}$.

Left-Hand and Centered Approximations

- There is nothing special about data to the **right**, why not the **left**?
- Again, Calculus gives us the **left-hand approximation**

$$D_- u(\bar{x}) := \frac{u(\bar{x}) - u(\bar{x} - h)}{h}, \quad h \ll 1.$$

- As before, in the limit $h \rightarrow 0$, this gives the **exact** derivative.
- As we shall see, both D_+ and D_- give answers which converge **linearly** in h .
- We can do **better** by averaging to realize the **centered difference approximation**

$$D_0 u(\bar{x}) := \frac{u(\bar{x} + h) - u(\bar{x} - h)}{2h}, \quad h \ll 1.$$

- It turns out that this formula is more accurate in that it converges **quadratically** in h .

The Geometry of Left, Right, and Centered Approximations

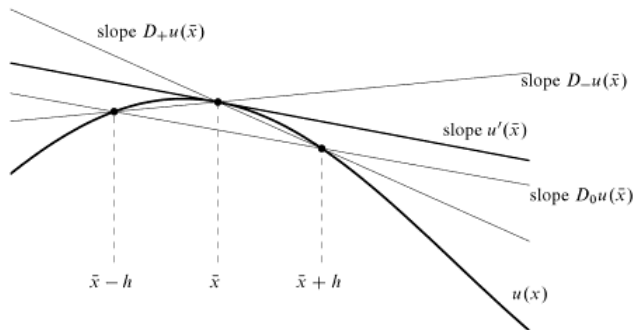


Figure 1.1. Various approximations to $u'(\bar{x})$ interpreted as the slope of secant lines.

A Third Order Approximation

- **Question:** Can we do better than these three approximations?
- **Answer:** Yes!
- For instance, one can consider this (uninspired) formula

$$D_3 u(\bar{x}) := \frac{2u(\bar{x} + h) + 3u(\bar{x}) - 6u(\bar{x} - h) + u(\bar{x} - 2h)}{6h}, \quad h \ll 1.$$

- This formula is even more accurate in that it converges **cubically** in h .

Example A: Let $u(x) = \sin(x)$ and $\bar{x} = 1$, and let us seek an approximation of $u'(1) = \cos(1) \approx 0.5403023$. The table below shows the error

$$Du(\bar{x}) - u'(\bar{x}),$$

for various choices of D and h .

Convergence of Left, Right, Centered, and Order Three Approximations

Table 1.1. *Errors in various finite difference approximations to $u'(\bar{x})$.*

h	$D_+u(\bar{x})$	$D_-u(\bar{x})$	$D_0u(\bar{x})$	$D_3u(\bar{x})$
1.0e-01	-4.2939e-02	4.1138e-02	-9.0005e-04	6.8207e-05
5.0e-02	-2.1257e-02	2.0807e-02	-2.2510e-04	8.6491e-06
1.0e-02	-4.2163e-03	4.1983e-03	-9.0050e-06	6.9941e-08
5.0e-03	-2.1059e-03	2.1014e-03	-2.2513e-06	8.7540e-09
1.0e-03	-4.2083e-04	4.2065e-04	-9.0050e-08	6.9979e-11

Solution A and Experimental Determination of Order of Accuracy

Solution A: One can show that

$$D_+ u(\bar{x}) - u'(\bar{x}) \approx -0.42h, \quad D_0 u(\bar{x}) - u'(\bar{x}) \approx -0.09h^2, \\ D_3 u(\bar{x}) - u'(\bar{x}) \approx 0.007h^3.$$

In the Figure on the following slide we plot these approximations on a **log-log scale**. We do this since an **error function**

$$E(h) \approx Ch^p,$$

satisfies

$$\log |E(h)| \approx \log |C| + p \log h,$$

so that these **power laws** are **linear** on log-log scales.

Errors in $Du(\bar{x})$

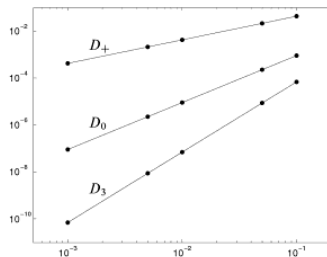


Figure 1.2. The errors in $Du(\bar{x})$ from Table 1.1 plotted against h on a log-log scale.

Appendix A: Measuring Errors

- While none of the material in this Appendix should be new or surprising, it is good to collect it all up for our use.
- Consider a problem in which the answer is a **single** value $\hat{z} \in \mathbf{R}$.
- For instance, perhaps we wish to approximate the solution $u(T)$ of

$$u'(t) = f(u(t)), \quad u(0) = \eta.$$

- If the approximated value is z then the **error** is

$$E := z - \hat{z}.$$

A.1: Errors in a Scalar Value

- The most natural measure of this error is the **absolute error**

$$|E| = |z - \hat{z}|.$$

- This works well when $\hat{z}$ is on the scale 10 – 10^{-1} .
- However, if $\hat{z}$ is **large** (e.g., 10^6) or **small** (e.g., 10^{-8}) this can be very **misleading**.
- For this reason one often uses the **relative error**

$$\left| \frac{z - \hat{z}}{z} \right|.$$

- Note:** Roughly speaking, if the **relative error** is 10^{-k} then k digits of z are **accurate**.
- Note:** While relative error “fixes” the problems with absolute error, it is usually best to **scale** your ODE/PDE so that z is 10 – 10^{-1} .

A.2: Big Oh Notation

An invaluable and easy-to-use (though precise) notation for **Order of Accuracy** is “**Big Oh**”.

Definition 1: If $f(h)$ and $g(h)$ are two functions of h , then we say that

$$f(h) = \mathcal{O}(g(h)) \quad \text{as } h \rightarrow 0$$

if there is some constant C such that

$$\left| \frac{f(h)}{g(h)} \right| < C, \quad h \text{ sufficiently small.}$$

Equivalently, if we can bound

$$|f(h)| < C |g(h)|, \quad h \text{ sufficiently small.}$$

Remark 2: Intuitively this means that f decays to zero **at least as fast as** the function g does.

Remark 3: Usually $g(h)$ is some monomial, h^q , $q \in \mathbb{Z}$.

A.2: Little Oh Notation

Another notation for **Order of Accuracy** is “**Little Oh**”.

Definition 4: If $f(h)$ and $g(h)$ are two functions of h , then we say that

$$f(h) = o(g(h)) \quad \text{as } h \rightarrow 0$$

if

$$\left| \frac{f(h)}{g(h)} \right| \rightarrow 0, \quad h \rightarrow 0.$$

Remark 5: This is slightly stronger than “Big Oh” as this means that f decays to zero **faster** than g . So,

$$f(h) = o(g(h)) \implies f(h) = \mathcal{O}(g(h)),$$

but **not** the converse!

Remark 6: $f(h) = o(1)$ means that $f(h) \rightarrow 0$ as $h \rightarrow 0$.

Big/Little Oh Examples

$$2h^3 = \mathcal{O}(h^3),$$

$$2h^3 = \mathcal{O}(h^2),$$

$$2h^3 = o(h^2),$$

$$\sin h = \mathcal{O}(h),$$

$$\sin h = h + o(h),$$

$$\sqrt{h} = \mathcal{O}(1), o(1),$$

$$1 - \cos h = \mathcal{O}(h^2),$$

$$2h^3/h^3 = 2,$$

$$2h^3/h^2 = 2h < 1, \text{ if } h < 1/2,$$

$$2h^3/h^2 = 2h \rightarrow 0,$$

$$\sin h = h - h^3/3! + \dots,$$

$$(\sin h - h)/h = \mathcal{O}(h^2),$$

$$\text{but } \sqrt{h} \neq \mathcal{O}(h),$$

$$\cos h = 1 - h^2/2! + \dots$$

1.1: Truncation Errors

- The **procedure** for analyzing errors in Finite Difference methods is to expand the function values of u in a **Taylor series** about $\bar{x}$, e.g.,

$$u(\bar{x} + h) = u(\bar{x}) + hu'(\bar{x}) + \frac{1}{2}h^2u''(\bar{x}) + \frac{1}{6}h^3u'''(\bar{x}) + \mathcal{O}(h^4),$$

$$u(\bar{x} - h) = u(\bar{x}) - hu'(\bar{x}) + \frac{1}{2}h^2u''(\bar{x}) - \frac{1}{6}h^3u'''(\bar{x}) + \mathcal{O}(h^4).$$

- From the definition of D_+ we have

$$D_+u(\bar{x}) = \frac{u(\bar{x} + h) - u(\bar{x})}{h} = u'(\bar{x}) + \frac{1}{2}hu''(\bar{x}) + \frac{1}{6}h^2u'''(\bar{x}) + \mathcal{O}(h^3).$$

- We note that $u^{(m)}(\bar{x})$ are **constants** as h is varied.
- Furthermore, for h small the error is dominated by $(h/2)u''(\bar{x})$.
- In the previous example $(1/2)u''(1) \approx -0.4207355$ which agrees with our simulations.

Truncation Error: D_0

- By combining the expansions for $u(\bar{x} + h)$ and $u(\bar{x} - h)$ we have

$$u(\bar{x} + h) - u(\bar{x} - h) = 2hu'(\bar{x}) + \frac{1}{3}h^3u'''(\bar{x}) + \mathcal{O}(h^5).$$

- From this we can demonstrate that

$$D_0u(\bar{x}) - u'(\bar{x}) = \frac{1}{6}h^2u'''(\bar{x}) + \mathcal{O}(h^4).$$

- The error in D_0 dominated by the **quadratic** term as h gets small.
- Again, regarding our example from before,

$$\frac{1}{6}u'''(\bar{x}) = -\frac{1}{6}\cos(1) \approx -0.09005038.$$

- Note:** All of the **odd** terms disappear in the error term. This is typical for **symmetric** approximations such as this one.

Truncation Error: D_3

- To analyze D_3 we require the **Taylor series**

$$u(\bar{x}-2h) = u(\bar{x}) - 2hu'(\bar{x}) + \frac{1}{2}(2h)^2u''(\bar{x}) - \frac{1}{6}(2h)^3u'''(\bar{x}) + \mathcal{O}(h^4).$$

- With this one can show that

$$D_3u(\bar{x}) = u'(\bar{x}) + \frac{1}{12}h^3u^{(4)}(\bar{x}) + \mathcal{O}(h^4).$$

Question: While D_+ , D_- , D_0 are “natural” from Calculus, D_3 drops from the sky. How does one derive D_3 ? Even better formulas?

1.2: Deriving Finite Difference Approximations

Happily there is a well-defined **procedure** for finding Finite Difference approximations of **arbitrary** order to derivatives of **arbitrary** order ... **provided** one samples at enough points.

Example B: Produce an approximation to $u'(\bar{x})$ using the points $u(\bar{x})$, $u(\bar{x} + h)$, and $u(\bar{x} + 3h)$.

Solution B: To begin we recall the Taylor series expansions

$$\begin{aligned}u(\bar{x} + h) &= u(\bar{x}) + hu'(\bar{x}) + \frac{1}{2}h^2u''(\bar{x}) + \mathcal{O}(h^3) \\u(\bar{x} + 3h) &= u(\bar{x}) + 3hu'(\bar{x}) + \frac{1}{2}(3h)^2u''(\bar{x}) + \mathcal{O}(h^3).\end{aligned}$$

Solution B

We now seek

$$\tilde{D}u(\bar{x}) = au(\bar{x}) + bu(\bar{x} + h) + cu(\bar{x} + 3h) \approx u'(\bar{x}),$$

for some choice of a, b, c . In order for this to be true we require that

$$a + b + c = 0,$$

$$bh + c(3h) = 1,$$

$$b(h^2/2) + c((3h)^2/2) = 0.$$

Solving this linear system of equations gives

$$a = -\frac{4}{3h}, \quad b = \frac{3}{2h}, \quad c = -\frac{1}{6h},$$

and

$$\tilde{D}u(\bar{x}) := \frac{-8u(\bar{x}) + 9u(\bar{x} + h) - u(\bar{x} + 3h)}{6h}.$$

Another Perspective

- Another approach to the problem of finding approximations to derivatives is to
 - 1 Find the **unique** polynomial, $p(x)$, which **interpolates** the function $u(x)$ at the points in question.
 - 2 **Differentiate** the function $p(x)$.
 - 3 **Evaluate** $p'(\bar{x})$.
- Happily, you will find that this gives the **same** answer as our Taylor series approach.
- This philosophy is very important in deriving other schemes for approximating solutions of ODEs/PDEs, particularly **Spectral Methods**.

Appendix A: Measuring Error

- Last time we discussed errors in a **scalar** value (both **absolute** and **relative**).
- In addition we carefully defined the “**Big Oh**” and “**Little Oh**” notation (which will be used frequently in this and other courses).
- Today we discuss some further measures of error which will be utilized in this course which may, depending on your background, be much less familiar.
- The issue is that we need to be able to measure the error in
 - **Vectors.**
 - **Functions.**
 - **Approximations of Functions.**
- We will give a full discussion before returning to the material of Chapter 1 on **Finite Differences**.

A.3: Errors in Vectors

- Suppose now that the quantity of interest is a **vector**, $\hat{z} \in \mathbf{R}^s$.
- For instance, perhaps we wish to approximate the solution $u(T)$ of a **system** of ODEs

$$u'(t) = f(u(t)), \quad u(0) = \eta.$$

- If the approximated value is $z \in \mathbf{R}^s$ then the **error** is

$$e := z - \hat{z} \in \mathbf{R}^s.$$

- We now measure this in a **norm** from **Linear Algebra**.
- Recall, there are **many** choices!

Vector Norms

Definition 1: A (vector) **norm** is a mapping, $\|\cdot\|$, from $\mathbf{R}^s$ to the nonnegative real numbers satisfying

- 1 $\|x\| \geq 0$ for any $x \in \mathbf{R}^s$, and $\|x\| = 0$ **if and only if** $x = 0$.
- 2 If $a \in \mathbf{R}$ is any scalar then

$$\|ax\| = |a| \|x\| .$$

- 3 If $x, y \in \mathbf{R}^s$ then the **triangle inequality** holds

$$\|x + y\| \leq \|x\| + \|y\| .$$

Definition 2: The ℓ^∞ **norm** is defined by

$$\|x\|_\infty := \max_{1 \leq j \leq s} |x_j| .$$

The ℓ^q Norms

Definition 3: For any $1 \leq q < \infty$ the ℓ^q norm is defined by

$$\|x\|_q := \left(\sum_{j=1}^s |x_j|^q \right)^{1/q}.$$

Remark 4: Two important special cases are $q = 1$ and $q = 2$ (Euclidean distance).

Remark 5: Typically one defines the ℓ^p spaces, but p is used for Order of Accuracy in Numerical Analysis.

A.3.1: Norm Equivalence

- **Question:** Do our results depend on the norm used?
- For instance, if $\|e(h)\|_2 = \mathcal{O}(h^p)$ then does $\|e(h)\|_\infty = \mathcal{O}(h^p)$?
- **Answer:** It depends on the **dimension** of the vector space.

Definition 6: Let $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ represent two different norms on $\mathbf{R}^s$. The norms are **equivalent** if there exist two positive constants, $C_1, C_2 \in \mathbf{R}^+$, such that

$$C_1 \|x\|_\alpha \leq \|x\|_\beta \leq C_2 \|x\|_\alpha,$$

for **any** $x \in \mathbf{R}^s$.

Theorem 7: All norms on $\mathbf{R}^s$ are **equivalent**.

Remark 8: It can be shown that

$$\begin{aligned} \|x\|_\infty \leq \|x\|_1 \leq s \|x\|_\infty, \quad \|x\|_\infty \leq \|x\|_2 \leq \sqrt{s} \|x\|_\infty, \\ \|x\|_2 \leq \|x\|_1 \leq \sqrt{s} \|x\|_2. \end{aligned}$$

A.3.2: Matrix Norms

Definition 9: For any vector norm $\|\cdot\|$ the **matrix norm** of a matrix $A \in \mathbf{R}^{s \times s}$ is the **smallest** $C \in \mathbf{R}$ such that

$$\|Ax\| \leq C \|x\|, \quad \forall x \in \mathbf{R}^s.$$

Equivalently,

$$\|A\| = \max_{x \in \mathbf{R}^s, x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{x \in \mathbf{R}^s, \|x\|=1} \|Ax\|.$$

Remark 10: Some convenient formulas

$$\|A\|_1 = \max_{1 \leq k \leq s} \sum_{j=1}^s |A_{j,k}| \quad \text{maximum column sum,}$$

$$\|A\|_\infty = \max_{1 \leq j \leq s} \sum_{k=1}^s |A_{j,k}| \quad \text{maximum row sum.}$$

The ℓ^2 Norm of a Matrix

Definition 11: The **spectral radius**, $\rho(A)$, of a matrix, $A \in \mathbf{R}^{s \times s}$, is the maximum of the moduli of the eigenvalues of the matrix.

Remark 12: It can be shown that

$$\|A\|_2 = \sqrt{\rho(A^T A)}.$$

Definition 13: The **condition number** of a matrix $A \in \mathbf{R}^{s \times s}$ is

$$\kappa(A) := \|A\| \left\| A^{-1} \right\|.$$

Remark 14: If A is **symmetric** then

$$\|A\|_2 = \rho(A), \quad \kappa_2(A) := \|A\|_2 \left\| A^{-1} \right\|_2 = \frac{\max |\lambda_j|}{\min |\lambda_j|},$$

where λ_j are the eigenvalues of A .

A.4: Errors in Functions

- Suppose now that the quantity of interest is a **function** $u : [a, b] \rightarrow \mathbf{R}$.
- As we shall see, the **Finite Element Method (FEM)** generates an approximate solution which is also a function so these developments will be useful later.
- If the approximate function is $U : [a, b] \rightarrow \mathbf{R}$ then the **error** is

$$e(x) := U(x) - u(x).$$

- We now measure this in a **norm** from **Functional Analysis**.
- As before, there are **many** choices!

Function Norms

Remark 15: Happily our notion of **norm** was sufficiently generic that x in **any** vector space works!

Definition 16: The L^∞ **norm** is defined by

$$\|u\|_\infty := \sup_{a \leq x \leq b} |u(x)|.$$

Definition 17: For any $1 \leq q < \infty$ the L^q **norm** is defined by

$$\|u\|_q := \left(\int_a^b |u(x)|^q dx \right)^{1/q}.$$

Remark 18: There are **many** details (e.g., using the **Lebesgue** rather than **Riemann** integral) which I am skipping over. Please see graduate **Analysis** and **Functional Analysis** courses for full details!

A.5: Errors in Grid Functions

- Unlike the **FEM**, the **Finite Difference Method (FDM)** does **not** produce a **function** as an approximation.
- The FDM produces a **vector**, e.g.,

$$U = (U_1, U_2, \dots, U_N)^T \in \mathbf{R}^N, \quad U_j \approx u(x_j).$$

- However, as we **refine** our calculation (e.g., by choosing a gridspacing h smaller) N **increases** since $N = \mathcal{O}(1/h)$!
- Measuring the error of our approximation in ℓ^∞ is fine.
- However, it is possible that the ℓ^2 norm of the error **grows** as N increases even if $U_j \rightarrow u(x_j)$!
- **The Problem:** The vector norms should be **correctly scaled**.

Grid Functions and Their Norms

Definition 19: Given a collection of gridpoints $\{x_j\}_{j=1}^N$ and a function $u(x)$, a **gridfunction** is a vector in $\mathbf{R}^N$ such that

$$U_j \approx u(x_j), \quad 1 \leq j \leq N.$$

Definition 20: The **error** in a gridfunction approximation is a vector in $\mathbf{R}^N$ with components

$$e_j = U_j - u(x_j), \quad 1 \leq j \leq N.$$

Definition 21: The ℓ^∞ norm of a gridfunction is

$$\|U\|_\infty := \max_{1 \leq j \leq N} |U_j|.$$

For any $1 \leq q < \infty$ the ℓ^q norm of a gridfunction is

$$\|U\|_q = \|U\|_{q,h} := \left(h \sum_{j=1}^N |U_j|^q \right)^{1/q} \approx \|u\|_q.$$

Norm Equivalence

- **Question:** For a gridfunction are all norms equivalent?
- **Answer:** It's complicated . . .
- On the one hand, for N **fixed** the answer is **yes**.
- However, the **point** is to send $N \rightarrow \infty$.
- In fact the answer is **no** so one must interpret results carefully.

Remark 22: It can be shown that

$$\begin{aligned}h \|U\|_{\infty} &\leq \|U\|_1 \leq (Nh) \|U\|_{\infty} = (b-a) \|U\|_{\infty}, \\ \sqrt{h} \|U\|_{\infty} &\leq \|U\|_2 \leq \sqrt{Nh} \|U\|_{\infty} = \sqrt{b-a} \|U\|_{\infty}, \\ \sqrt{h} \|U\|_2 &\leq \|U\|_1 \leq \sqrt{Nh} \|U\|_2 = \sqrt{b-a} \|U\|_2.\end{aligned}$$

Example A

Example A: Consider the step function

$$u(x) = \begin{cases} 0, & x \leq 1/2, \\ 1, & x > 1/2, \end{cases}$$

and the gridfunction

$$U = \begin{cases} 0, & x_j < 1/2, \\ 1/2, & x_j = 1/2, \\ 1, & x_j > 1/2. \end{cases}$$

Does the gridfunction converge to the function?

Solution A: It can be shown that

$$\|e(h)\|_{\infty} = 1/2, \quad \|e(h)\|_1 = h/2 = \mathcal{O}(h), \quad \text{as } h \rightarrow 0,$$

so “no” and “yes!”

A.6: Estimating Errors in Numerical Solutions

- A crucial consideration in the numerical solution of ODEs/PDEs is “How well are we doing?”
- Suppose that $U(h) \in \mathbf{R}^N$ is our approximate solution.
- If we have the **exact solution**, $u(x)$, then this is straightforward: For $\hat{U}_j = u(x_j)$, simply study

$$E(h) = \|U(h) - \hat{U}(h)\|.$$

- If the method is **Order- p** then we expect

$$E(h) = Ch^p + o(h^p) \quad \text{and} \quad E(h) \approx Ch^p.$$

- Thus, if $R(h) := E(h)/E(h/2)$, then refining by a factor of **two**,

$$E(h/2) \approx C(h/2)^p \quad \implies \quad R(h) \approx 2^p.$$

- From this we conclude that $p \approx \log_2 R(h)$.

Estimates from Finer Solutions

- Of course if we had an exact solution we would not be using a numerical approximation in the first place! What to do?
- Suppose we complete a simulation on a very **fine** mesh with spacing $\bar{h}$.
- Provided that the fine mesh points are in the coarse mesh define $\bar{U}(h)$ as the **fine mesh solution on the coarse mesh**.
- Then

$$\begin{aligned}\bar{E}(h) &:= \|U(h) - \bar{U}(h)\| = \|U(h) - \hat{U}(h) + \hat{U}(h) - \bar{U}(h)\| \\ &\leq E(h) + \|\hat{U}(h) - \bar{U}(h)\|.\end{aligned}$$

- If $\bar{h}^p \ll h^p$ then the second term is negligible and $\bar{E}(h)$ gives a good estimate.
- Now we can proceed as if we had an exact solution.

1.3: Second Order Derivatives

- So far we have only considered the **first derivative** of a function, but higher derivatives are important as well.
- The standard **centered approximation** is

$$\begin{aligned} D^2 u(\bar{x}) &:= \frac{u(\bar{x} - h) - 2u(\bar{x}) + u(\bar{x} + h)}{h^2} \\ &= u''(\bar{x}) + \frac{1}{12}h^2 u^{(4)}(\bar{x}) + \mathcal{O}(h^4). \end{aligned}$$

- This can be derived in many ways and a natural one is to apply **two** first derivative approximations

$$D^2 u(\bar{x}) = D_+ D_- u(\bar{x}) = D_- D_+ u(\bar{x}) = \hat{D}_0 \hat{D}_0 u(\bar{x}),$$

where

$$\hat{D}_0 u(x) = \frac{u(\bar{x} + h/2) - u(\bar{x} - h/2)}{h}.$$

Example B

Example B: Approximate $u''(x_2)$ using

$$U_j = u(x_j), \quad j = 1, 2, 3.$$

Solution B: Define

$$h_1 = x_2 - x_1, \quad h_2 = x_3 - x_2,$$

so that $x_3 - x_1 = h_1 + h_2$. The **Newton form** (see Appendix B) of the **interpolating polynomial** is

$$p(x) = a + b(x - x_1) + c(x - x_1)(x - x_2).$$

Evaluating this at $x = x_1$ gives

$$a = U_1 =: U[x_1].$$

Evaluating at $x = x_2$ together with a gives

$$b = \frac{U_2 - U_1}{h_1} =: U[x_1, x_2].$$

Solution B, cont.

Finally, evaluating at $x = x_3$ coupled with a and b delivers

$$c = \frac{(U_3 - U_2)/h_2 - (U_2 - U_1)/h_1}{h_1 + h_2} =: U[x_1, x_2, x_3].$$

Now,

$$p''(x_2) = 2U[x_1, x_2, x_3] = c_1 U_1 + c_2 U_2 + c_3 U_3,$$

where

$$c_1 = \frac{2}{h_1(h_1 + h_2)}, \quad c_2 = -\frac{2}{h_1 h_2}, \quad c_3 = \frac{2}{h_2(h_1 + h_2)}.$$

1.4: Higher Order Derivatives

- There are many ways to derive approximations of higher order derivatives using the methods we have outlined.
- For instance,

$$\begin{aligned}D_+ D^2 u(\bar{x}) &= \frac{u(\bar{x} + 2h) - 3u(\bar{x} + h) + 3u(\bar{x}) - u(\bar{x} - h)}{h^3} \\&= u'''(\bar{x}) + \frac{1}{2} h u^{(4)}(\bar{x}) + \mathcal{O}(h^2).\end{aligned}$$

- Alternatively,

$$\begin{aligned}D_0 D_+ D_- u(\bar{x}) &= \frac{u(\bar{x} + 2h) - 2u(\bar{x} + h) + 2u(\bar{x} - h) - u(\bar{x} - 2h)}{2h^3} \\&= u'''(\bar{x}) + \frac{1}{4} h^2 u^{(5)}(\bar{x}) + \mathcal{O}(h^4).\end{aligned}$$

1.5: A General Approach to Deriving the Coefficients

- The method of § 1.2 can be generalized and systematized into the **Method of Undetermined Coefficients**.
- This method delivers an approximation to $u^{(k)}(\bar{x})$ based on an **arbitrary stencil** of $n \geq k + 1$ points, $\{x_1, \dots, x_n\}$.
- The **Taylor series** we require are

$$u(x_j) = u(\bar{x}) + (x_j - \bar{x})u'(\bar{x}) + \dots + \frac{1}{k!}(x_j - \bar{x})^k u^{(k)}(\bar{x}) + \dots$$

- We desire that, for p **as large as possible**,

$$c_1 u(x_1) + \dots + c_n u(x_n) = u^{(k)}(\bar{x}) + \mathcal{O}(h^p).$$

Method of Undetermined Coefficients, cont.

- The approach of § 1.2 demands that

$$\frac{1}{(j-1)!} \sum_{\ell=1}^n c_{\ell} (x_j - \bar{x})^{\ell-1} = \begin{cases} 1, & j-1 = k, \\ 0, & \text{otherwise.} \end{cases}$$

- If the points are distinct this **Vandermonde system** is nonsingular.
- The order of accuracy satisfies $p \geq n - k$.

MATLAB code for Finite Difference Coefficients

```
function c = fdcoeffV(k,xbar,x)
A = ones(n,n);
xrow = (x(:)-xbar)'; % displacements as a row vector.
for i=2:n
    A(i,:) = (xrow .^ (i-1)) ./ factorial(i-1);
end
b = zeros(n,1);      % b is right hand side,
b(k+1) = 1;          % so k'th derivative term remains
c = A\b;              % solve system for coefficients
c = c';               % row vector
```